

6

House of Hundreds - I



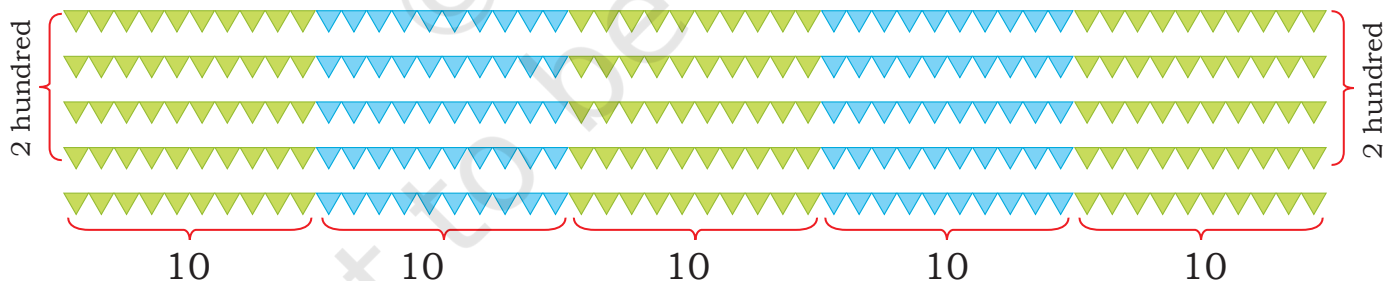
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Ajji, Teji and Jojo have come to the mela. Guess the number of 'triangular torans'.



Now, find out how many there are.



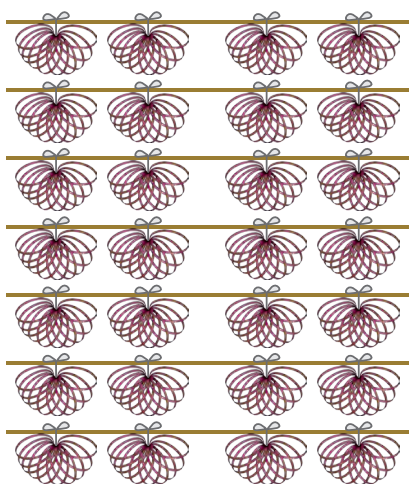
How many triangles are there in a line of triangular toran?

Try counting all the triangles.

Total triangle: 50 more than 200, which is 250.



Teacher's Note: Help children to count objects beyond 200 and show how the counting can be done as earlier using the same number names.



Guess how many bangles there are.

Try counting the total number of bangles.

See how Teji is counting.

Total bangles: 200 and 80 more is 280.

Bangles! Let's count.

10, 20, 30, ... 100, 110,
120, ... 200, 210, ... 270,
271, 272, ... 279, 280

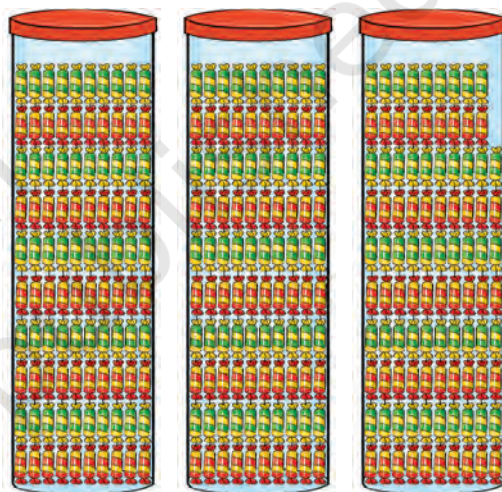


Guess how many toffees there are in the boxes.

Count and check.



These are lots of toffees!
Let's count. 10, 20, 30, ...
100, 110, 120, ... 190, 200,
210, ... 290, 291, 292,
293, ... 298



Jojo has 2 toffees in his hand.

How many toffees are there altogether?

298 and
one more is 299;
299 and one
more is 300.

$$298 + 1 = 299$$

$$299 + 1 = 300$$

How many more triangles to make 300?

How many bangles less than 300?

Which is more: bangles or triangles?



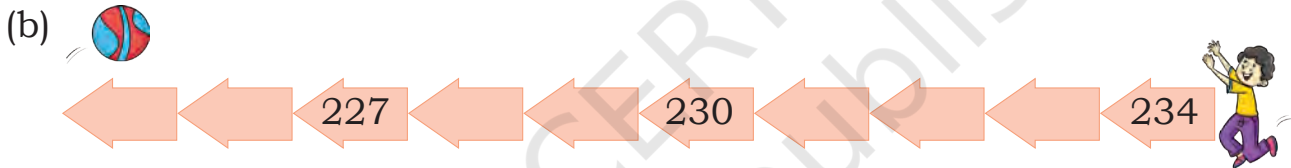


Let us Do

1. Jojo is jumping on a tiled path inside the mela.
Fill in the empty tiles with numbers.



2. Fill in the blanks with the correct numbers.



3. Ants have found food on the ground. Guess how many ants there are. Count and check.



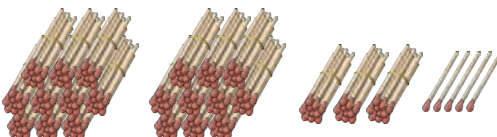
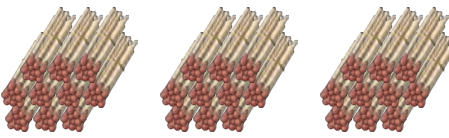
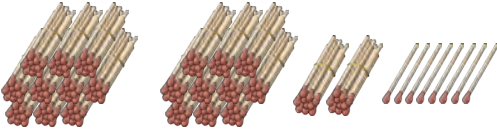
Teacher's Note: Ask why children got different answers and how one can get better at counting. The differences in the answers should be used as an opportunity to show why groups of 10 are more effective in counting correctly even large numbers. Help children arrive at a strategy to count correctly.



Teji and Jojo have learnt to write numbers with the help of matchstick bundles. They can also write number sentences in more than one way.



4. Fill in the blanks appropriately.

Matchsticks	Number	Number Sentence
	235	200 and 35 more ($200 + 35$) 15 less than 250 ($250 - 15$)
		
		
		300 and 16 more
	109	

5. Place the numbers given above on the number line.

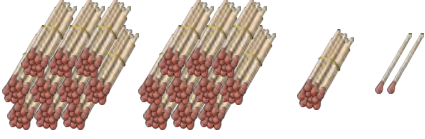
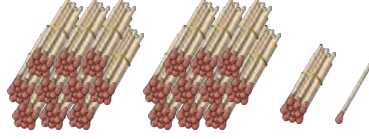
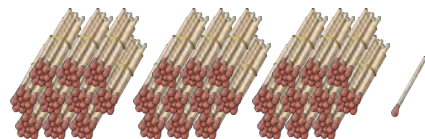
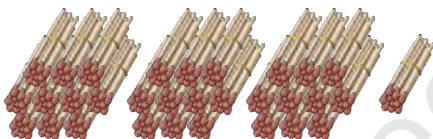
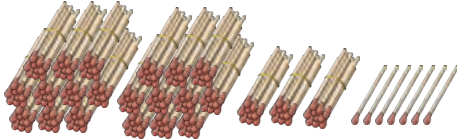
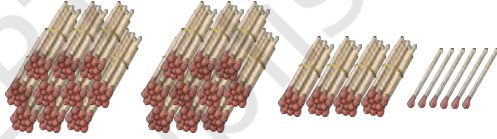


235 lies between 200 and 250.



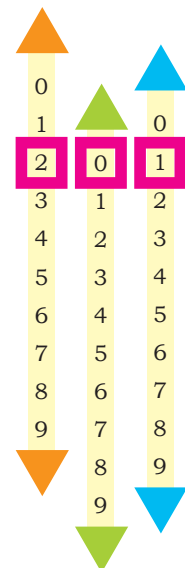
Teacher's Note: Ask children to make large numbers using matchsticks or any other readily available material at home and bring to school.

6. Look at the pictures and write the corresponding numbers.

7. Make the number slider as shown in the picture. Increase or decrease the number as given below:

- (a) 285—increase the number by one
- (b) 147—increase the number by ten
- (c) 367—decrease the number by 2
- (d) 289—decrease the number by 10
- (e) 290—increase the number by 20

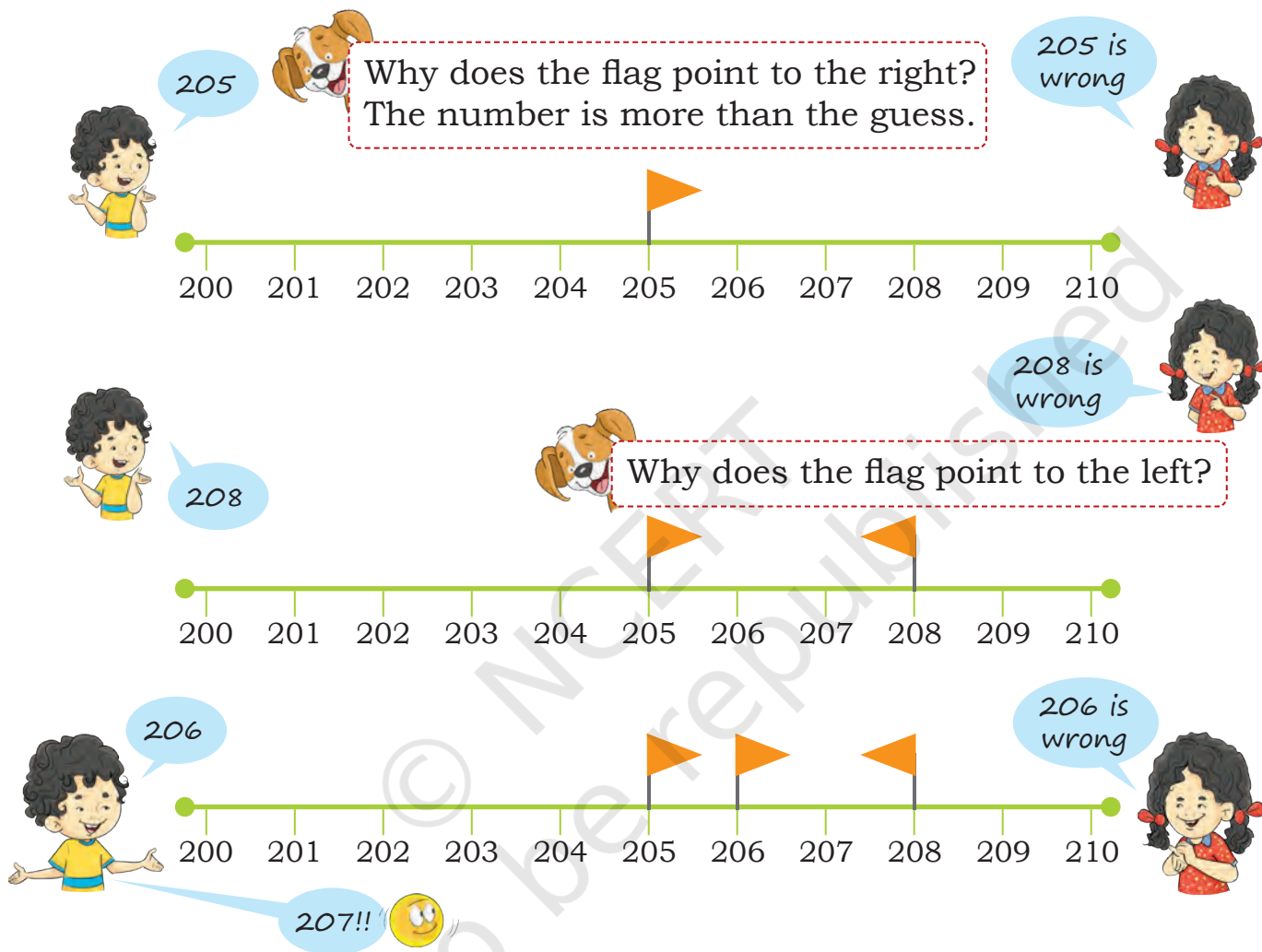


Teacher's Note: Teacher to make the TLM and play with children.



Let us Play

Flag game: Let us play a guessing game. Teji has thought of a number between 200 and 210. Jojo has to guess it.



Now, try and play with your friends, and guess the number.



Teacher's Note: Play this game on the board with children; later, children can play in pairs or small groups in their notebooks. The teacher can also make the game more exciting by changing the range of the numbers and restricting the number of guesses allowed.

Magical Count

Write down any number name. Count the number of letters in that number name and write the name of that new number down. Keep repeating—what happens?

235 Two hundred and thirty five (23 letters)
Twenty-three (6 letters)
Eleven (5 letters)
Six (4 letters)
Three (3 letters)
Five (11 letters)
Four (11 letters)

187 One hundred and eighty seven

204 Two hundred and four

Will it always end with **FOUR**?

Numbers on a line

Tell Teji, Jojo and Bholu what will be the next hundred number. Write it on the number line below.

Can you show the number using matchsticks?



One hundred,
...two hundred,
...three hundred,
...and ???
FOUR HUNDRED



Let us Do

Teji and Jojo are trying to put their numbers on the following number lines.

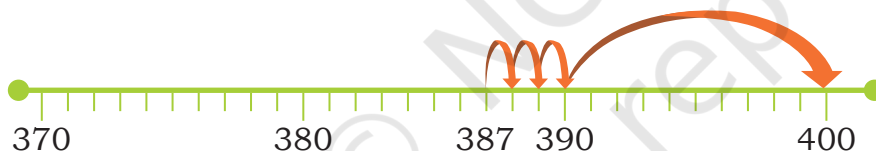
1. Locate 216, 243, 257 on the number line below.



2. Locate 329, 332, 337, 375 and 387 on the number line below.



3. Tell how far is 387 from 400:



4. Which is more: 393 or 400? Use a number line and show. Fill the numbers on the number line below and show by jumping how far 393 is from 400.

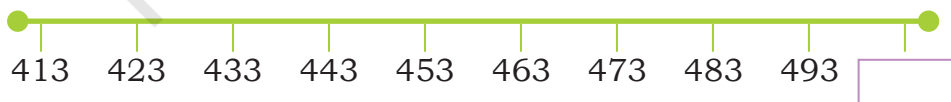


Teacher's Note: Support children to pay attention to the different jumps required in the above number lines. Teacher to also support children to understand quantities that the numbers represent using matchstick bundles or any other similar material.

5. Teji and Jojo are hungry after all the running around.
They go to Farooq Chacha's sweet shop.



- (a) How many pieces of *mysore pak* are in one tray?
- (b) How many pieces of *mysore pak* are there in total?
- (c) How many laddoos does chacha have in the trays?
- (d) How many *dhoklas* does chacha have?
- (e) Chacha is going to fill the tray with more laddoos. How many more laddoos will make the tray full?
- (f) How many total laddoos will he have after the last tray is full?
- (g) Mark the following numbers on the number line below:
423, 487, 438, 476.



Mark Five
Hundred
(500) on this
number line!



Let us Do

1. Teji and Jojo saw a big apartment.

Find the house numbers of the shaded houses without counting. Share your strategies.

One floor above 105 is 115.



Fill in the shaded boxes with the correct house numbers.



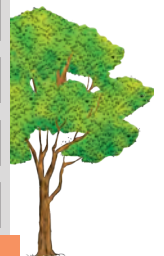
Column ↓

HOP HUNDREDS HOME

Row →

					186				
			174						
				155					160
		143							
							138		
						127			
111	112	113							
101	102	103	104	105	106	107	108	109	110
									100
11	12	13	14	15	16	17	18	19	20
01	02	03	04	05	06	07	08	09	10

113 is 10 floors above 13



Teacher's Note: Let children fill the first blank cell by counting, but encourage them to look for patterns while filling in the rest.



2. Arvind Dada has to deliver sweets from Farooq chacha's shop to different houses. Colour the houses to which he has to deliver sweets.

The house numbers are:

209, 228, 242, 258, 267, 276, 290, 315, 346, 367, 389, 395.

JUMP JOY RESIDENCY

Column

Row

301

242

211 212

201 209

1 floor above 201 is 211. I can reach 212 by going one house to the right.

3. Write the floor and column number for each of the following houses.

House Number	Floor	Column
13	1st	3
67		
106		
159		
192		
231		
245		
328		
380		
399		

4. Find the following house numbers from the building and write the appropriate house numbers in the blank spaces. What do you notice? Discuss how the house numbers change when moving up and down, and left to right.

					238
12					
	144				319

5. Who am I?

I have digits 9, 1 and 5.
I am less than 200.
I have 9 ones.
Which number am I?

I am a 3-digit number. I have only digits 4 and 0.
Which number am I?

I am greater than 300 but less than 400.
I have no tens.
My ones and hundreds digits are the same.
Which number am I?

Arvind Dada packs sweets in boxes of 100 (H), 10 (T) and as packets of loose sweets (O). The number of sweets for every house is the same as the house number.

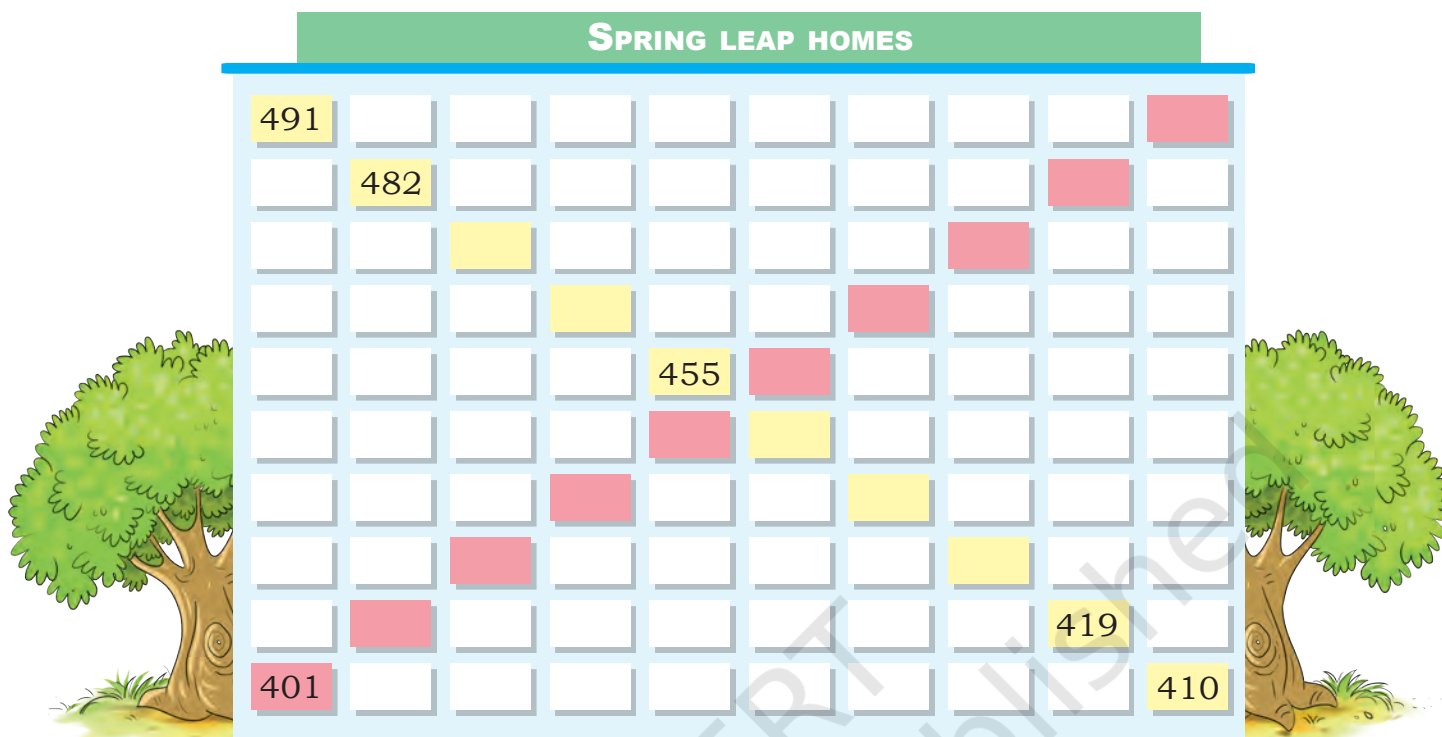
6. Draw sweets for each of the following house numbers.

House Numbers	Draw Sweet Boxes	Types of Boxes and Packets	Number Sentence
211	H  T  O 	2 H + 1 T + 1 O	200 + 10 + 1
309			
275			
423			
365			
343			
458			
562			
606			
800			



Teacher's Note: Children are supposed to show the sweets using Dienes blocks. One set is provided at the end of the book. These can be easily made using square math notebooks. Children can make a model using Dienes blocks before drawing.

7. (a) Write the house numbers of the yellow and pink houses.



- (b) Write the pattern you see in these numbers.

.....

.....

8. Arvind dada wants to pack small boxes of 10 in a big box of 100.

- How many boxes of 10 can he fit in a box of 100?
- How many boxes of 10 can he fit in two boxes of 100?
- How many boxes of 10 can he fit in four boxes of 100?
- How many boxes of 10 will he find if he opens a box of 100?
.....



Teacher's Note: Teacher can encourage children to identify patterns in the numbers, some of the digits, how the digits change, etc. Also, help children find the relationships among the 100's box, 10's box and 1's box.

9. Number hunt

Write the numbers between 200 and 300 that have 5 as a digit.
Is 245 one such number? Write the other numbers.

.....

.....



Let us Play

Do you remember this game? Let us play it again.
We will record the actions in the table below. One is done for you.



Hundreds	Tens	Ones	Number
Clap Clap	Snap	Pat	211

Show and Tell

Create a chain of cards such that every next card answers the question of the previous card. Distribute these cards among the children in the class. A child reads aloud their card and the other child having the answer identifies themselves. The game ends when every child has answered a question using their card.

One example is given below:

I have two hundred and sixteen.

Who has ten more than this?

I have

Who has 308?

I have three hundred and eight.

Who has ?

I have four hundreds, two tens and zero units.

Who has this?

Let us compare who has more laddoos and show it using the sign more than (>) or less than (<) appropriately.

487 laddoos is more than 423 laddoos.

423 laddoos is less than 487 laddoos.



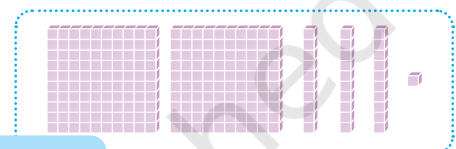
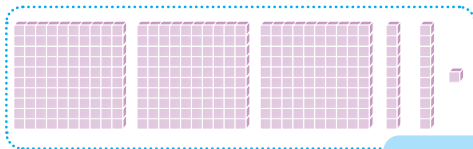
$$487 > 423$$

Open mouth points towards the bigger number



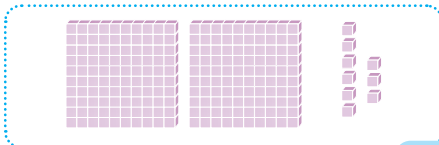
$$423 < 487$$

Now, compare 321 and 231. 3 hundreds are more than 2 hundreds. Do the other digits in the numbers matter here? No. So, 321 is more than (>) 231.



$$321 > 231$$

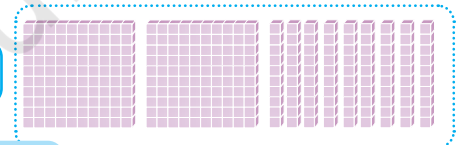
Let us take another example. We are comparing 209 with 290. Both numbers have two hundreds. But 9 ones are less than 9 tens. So, 209 is less than (<) 290.



209



290



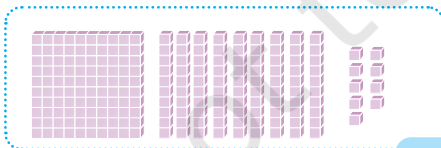
$$209 < 290$$



Let us Do

1. Compare the following numbers and use the signs >, < appropriately.

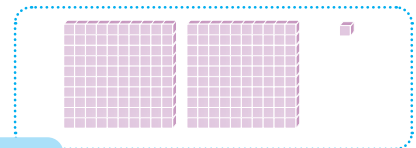
(a)



199

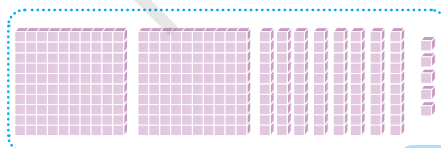


201



199 is than 201

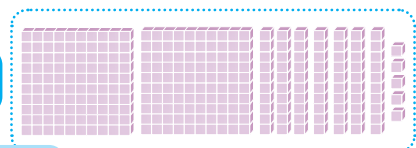
(b)



285



275



285 is than 275

2. Think and match the following.

(a) 325 is more than 235 because

329 and 392 both have three hundreds. 329 has 2 tens, 392 has 9 tens.

(b) 235 is less than 523 because

110 has 1 hundred and 11 has no hundreds (zero hundreds)

(c) 157 is more than 153 because

235 has 2 hundreds and 523 has 5 hundreds.

(d) 432 is more than 423 because

325 has 3 hundreds and 235 has 2 hundreds.

(e) 329 is less than 392 because

157 and 153 both have one hundred and 5 tens each. 157 has 7 ones, 153 has 3 ones.

(f) 110 is more than 11 because

432 and 423 both have four hundreds. 432 has 3 tens, 423 has 2 tens.

3. Circle the smallest number in each row:

(a) 374, 473, 347, 437

(b) 239, 123, 321, 456

4. Circle the greatest number in each row:

(a) 466, 437, 439, 447, 483

(b) 464, 387, 123, 256, 348

5. Make 3-digit numbers using 3, 2 and 4 without repeating any digit, and colour the greatest number with red and smallest number with yellow.

6. Now, make more 3-digit numbers using 3, 2 and 4 where you may repeat the digits. Colour the greatest number with red and smallest number with yellow.

7. (a) Arrange the following numbers from smallest to biggest.

456, 389, 207, 99, 110

.....

- (b) Arrange the following numbers from biggest to smallest.

67, 376, 294, 249, 494

.....



Teacher's Note: While comparing two numbers, help children focus on the quantities that the numbers represent. Use Dienes block representation to help them see that 1 H is more than 1 T and 1 O. Similarly, 1 T is more than 1 O.